Malaria Metapopulation Model

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Overview

Not all models in Starsim need to be agent-based. This vignette implements a **compartmental (ODE) malaria model** as a Starsim module, demonstrating how the framework supports non-agent-based components.

The model is a multi-patch **Ross-Macdonald** system with: - Seasonal mosquito density forcing (sinusoidal) - Human mobility between patches - Euler integration within the Starsim timestep loop

This follows `code/malaria_demo/demo_sim_ss.py`.

Helper functions

```julia
using Starsim
using LinearAlgebra
using Plots

function analytical_mosquito_density(ivector, hvector, pij, a, b, c, mu, r, tau)
    xvector = ivector ./ r
    gvector = xvector .* r ./ (1 .- xvector)
    n = length(xvector)
    fvector = zeros(n)
    for i in 1:n
        k_i = sum(pij[:, i] .* xvector .* hvector) / sum(pij[:, i] .* hvector)
        fvector[i] = b * c * k_i / (a * c * k_i / mu + 1)
    end
    fmatrix = Diagonal(fvector)
    cvector = (pij * fmatrix) \ gvector
    m = cvector .* mu ./ a^2 ./ exp(-mu * tau)
    return m
```

```julia
end

function seasonal_sinusoidal(t; amplitude=0.8, peak_day=180)
    day_of_year = t % 365
    return max(0.0, 1 + amplitude * sin(2  * (day_of_year - peak_day + 365/4) / 365))
end
```

```
seasonal_sinusoidal (generic function with 1 method)
```

Define the Malaria module

```julia
mutable struct MalariaODE <: AbstractModule
    mod::ModuleData

    # Bionomic parameters
    a::Float64      # Biting rate
    b::Float64      # Mosquito-to-human transmission
    c::Float64      # Human-to-mosquito transmission
    r::Float64      # Recovery rate
    mu::Float64     # Mosquito death rate
    tau::Float64    # Extrinsic incubation period

    # Patch data
    numpatch::Int
    hvector::Vector{Float64}   # Population sizes
    ivector::Vector{Float64}   # Initial incidence
    pij::Matrix{Float64}       # Mobility matrix
    m_base::Vector{Float64}    # Baseline mosquito density

    # Compartment state
    X::Vector{Float64}  # Prevalence per patch
    C::Vector{Float64}  # Cumulative incidence per patch
end

function MalariaODE(;
    name::Symbol = :malaria,
    a = 0.3, b = 0.1, c = 0.214,
    r = 1/150, mu = 1/10, tau = 10.0,
    hvector = [5000.0, 10000.0, 8000.0],
    ivector = [0.001, 0.003, 0.002],
```

```
    pij = [0.85 0.10 0.05;
           0.08 0.80 0.12;
           0.06 0.09 0.85],
)
    md = ModuleData(name; label="Malaria ODE")
    n = length(hvector)
    MalariaODE(md, a, b, c, r, mu, tau,
               n, copy(hvector), copy(ivector), copy(pij),
               zeros(n), zeros(n), zeros(n))
end

Starsim.module_data(m::MalariaODE) = m.mod

function Starsim.init_pre!(m::MalariaODE, sim)
    md = m.mod
    md.t = Timeline(start=sim.pars.start, stop=sim.pars.stop, dt=sim.pars.dt)

    npts = md.t.npts
    results = Result[]
    for i in 1:m.numpatch
        push!(results, Result(Symbol("X_$i"); npts=npts, label="Prevalence (Patch $i)", scale
        push!(results, Result(Symbol("C_$i"); npts=npts, label="Cum. incidence (Patch $i)", s
        push!(results, Result(Symbol("daily_inc_$i"); npts=npts, label="Daily incidence (Pat
        push!(results, Result(Symbol("m_seasonal_$i"); npts=npts, label="Mosquito density (Pa
        push!(results, Result(Symbol("eir_$i"); npts=npts, label="EIR (Patch $i)", scale=fals
    end
    define_results!(m, results...)

    md.initialized = true
    return m
end

function Starsim.init_post!(m::MalariaODE, sim)
    m.m_base = analytical_mosquito_density(
        m.ivector, m.hvector, m.pij,
        m.a, m.b, m.c, m.mu, m.r, m.tau,
    )
    m.m_base = max.(m.m_base, 1e-10)

    m.X .= m.ivector ./ m.r
    m.C .= 0.0
    return m
```

```julia
end

function Starsim.step!(m::MalariaODE, sim)
    md = m.mod
    t = md.t.start + (md.t.ti - 1) * md.t.dt
    dt = md.t.dt

    X = m.X
    H = m.hvector
    pij = m.pij
    n = m.numpatch

    # Seasonal mosquito density
    ms = m.m_base .* seasonal_sinusoidal(t)

    # Weighted prevalence at each destination
    k = (pij' * (X .* H)) ./ (pij' * H)

    # Force of infection
    Z_numer = m.a^2 * m.b * m.c * exp(-m.mu * m.tau) .* k
    Z_denom = m.a * m.c .* k .+ m.mu
    dC = pij * (ms .* Z_numer ./ Z_denom) .* (1 .- X)
    dX = dC .- m.r .* X

    # Euler update
    m.X .+= dX .* dt
    m.C .+= dC .* dt

    return m
end

function Starsim.update_results!(m::MalariaODE, sim)
    md = m.mod
    ti = md.t.ti
    t = md.t.start + (ti - 1) * md.t.dt
    n = m.numpatch

    for i in 1:n
        md.results[Symbol("X_$i")][ti] = m.X[i]
        md.results[Symbol("C_$i")][ti] = m.C[i]
    end
```

```
    # Derived quantities
    X = m.X; H = m.hvector; pij = m.pij
    ms = m.m_base .* seasonal_sinusoidal(t)
    k = (pij' * (X .* H)) ./ (pij' * H)

    Z_numer = m.a^2 * m.b * m.c * exp(-m.mu * m.tau) .* k
    Z_denom = m.a * m.c .* k .+ m.mu
    dC = pij * (ms .* Z_numer ./ Z_denom) .* (1 .- X)

    Z = m.a * m.c .* k .* exp(-m.mu * m.tau) ./ (m.a * m.c .* k .+ m.mu)
    eir = ms .* m.a .* (pij * Z)

    for i in 1:n
        md.results[Symbol("daily_inc_$i")][ti] = dC[i]
        md.results[Symbol("m_seasonal_$i")][ti] = ms[i]
        md.results[Symbol("eir_$i")][ti] = eir[i]
    end
    return m
end

function Starsim.finalize!(m::MalariaODE)
    return m
end
```

Running the model

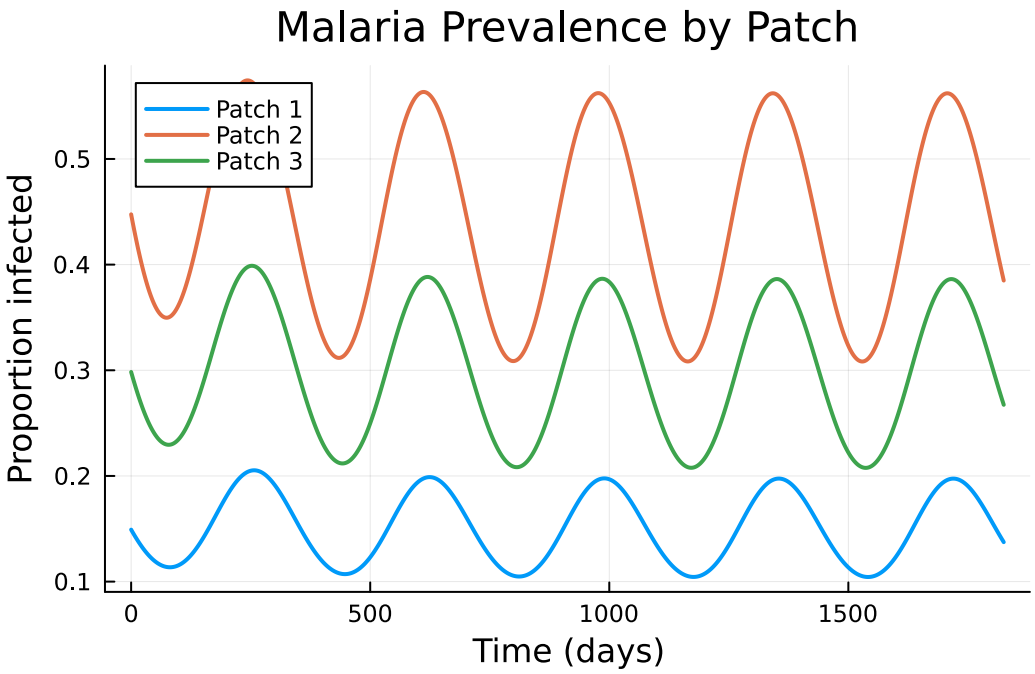
```
mal = MalariaODE()
sim = Sim(
    modules = mal,
    n_agents = 1,
    start = 0.0,
    stop = 365.0 * 5,
    dt = 1.0,
    verbose = 0,
)
run!(sim)
```

```
Sim(1 agents, 0.0→1825.0, dt=1.0, nets=0, dis=0, status=complete)
```

Prevalence dynamics

```
npts = sim.t.npts
tvec = 0:npts-1
patches = ["Patch 1", "Patch 2", "Patch 3"]

p = plot(xlabel="Time (days)", ylabel="Proportion infected",
         title="Malaria Prevalence by Patch")
for (i, name) in enumerate(patches)
    prev = get_result(sim, :malaria, Symbol("X_$i"))
    plot!(p, tvec, prev, lw=2, label=name)
end
p
```

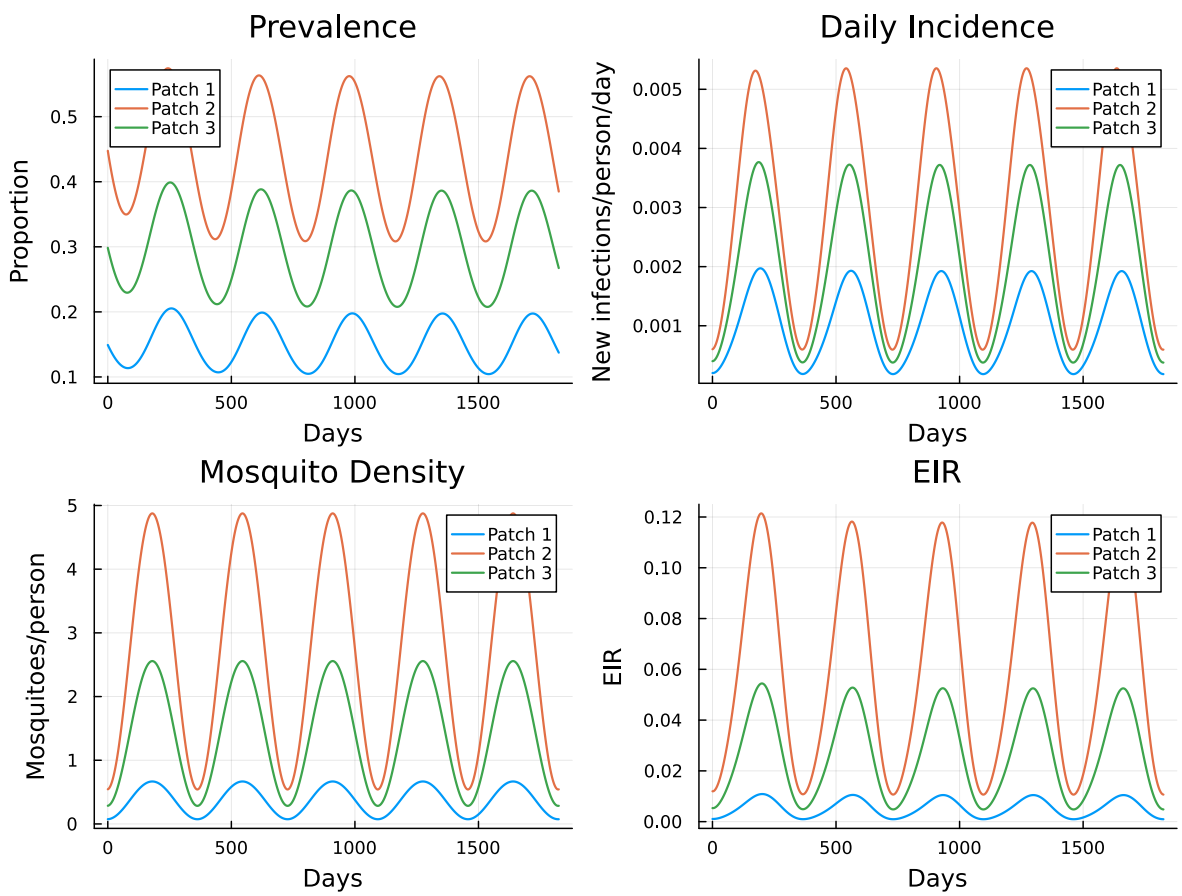


Four-panel dashboard

```
p1 = plot(title="Prevalence", xlabel="Days", ylabel="Proportion")
p2 = plot(title="Daily Incidence", xlabel="Days", ylabel="New infections/person/day")
p3 = plot(title="Mosquito Density", xlabel="Days", ylabel="Mosquitoes/person")
p4 = plot(title="EIR", xlabel="Days", ylabel="EIR")
```

```
for (i, name) in enumerate(patches)
    plot!(p1, tvec, get_result(sim, :malaria, Symbol("X_$i")), label=name, lw=1.5)
    plot!(p2, tvec, get_result(sim, :malaria, Symbol("daily_inc_$i")), label=name, lw=1.5)
    plot!(p3, tvec, get_result(sim, :malaria, Symbol("m_seasonal_$i")), label=name, lw=1.5)
    plot!(p4, tvec, get_result(sim, :malaria, Symbol("eir_$i")), label=name, lw=1.5)
end

plot(p1, p2, p3, p4, layout=(2, 2), size=(800, 600))
```



Summary

- Starsim supports non-agent-based modules via the generic `modules` keyword
- The Ross-Macdonald model uses Euler integration within the Starsim timestep loop
- Seasonal mosquito forcing creates annual cycles in prevalence, incidence, and EIR
- Multi-patch mobility allows infection to spread between populations

- This approach enables coupling ODE models with agent-based components in the same simulation